



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: V Semester B.E Computer Science and Engineering
Course Code & Title: CB3491 Cryptography and Cyber Security
Name of the Faculty member: Mrs. P.Sabeena Burvin, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit V / Classifications of Cyber Crimes
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO 10
- **Activity Chosen:** Learning by Teaching
- **Date Implementation:** 15.10.2025
- **Justification:**

The primary goal of the learning-by-teaching activity was to foster active learning and consolidate students' knowledge by having them teach assigned topics to their peers. This approach aimed to enhance their understanding of the material, as explaining concepts requires a deep comprehension and thorough review. Additionally, it sought to improve students' communication skills by having them articulate their knowledge clearly and effectively. The activity promoted active participation, transforming students from passive recipients to active contributors in the learning process, and fostered a collaborative environment where peers could learn from each other.

Time Allotted for the Activity: 30 minutes

Details of the Implementation:

- Prior to the session, the instructor asked the students about their willingness to teach a particular topic Classification of cybercrimes on 11.10.2025.
- Rishikesh R expressed interest in taking the class.
- Assigned the topic based on its relevance and complexity.
- The student was take 4 days' time to prepare the assigned topic and was encouraged to gather information from various sources, organize the content effectively, and plan her teaching approach.
- During the session, students took turns presenting their assigned topics to their classmates.
- After the presentations, a question-and-answer session was held to encourage active participation and clarify their doubts.
- This allowed students to engage in discussions and exchange additional insights related to the topic. Finally, the instructor summarized the content.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO8	PO12	PSO1
CO1	3	2	2	2	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO Mappe

Innovative Practice	PO1	PO2	PO3	PO4	PO8	PO12	PSO1
	3	2	2	2	1	1	1
Justification for correlation	Basic Computer Science Engineering and mathematics Knowledge for identifying various cybercrimes.	Students will be able to analyze and define various types of cybercrimes.	Students will be able to analyze and identify various cryptographic algorithms.	Students will be able to use the virtual environment for identifying cybercrimes.	Students will be able to use modern tools to perform identification process.	Students will work effectively as an individual and team for solving the identification process.	Updating the concepts to the current trends are moderately correlated.

• **Images / Screenshot of the practice:**



Figure 1: Discussion about Classification of Cyber Crimes by RISHIKESH R

Reflective Critique:**❖ *Feedback of practice from students and other stakeholders:***

- Presenting the seminar helped to understand the topic more deeply as it had to explain it to others.
- Students found the seminar activity to be a valuable learning experience that enhanced their understanding, communication skills, and confidence

❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- By involving students in teaching and discussions, the seminar promotes active learning, making the educational experience more dynamic and interactive.
- Students spent their time in self-learning.
- The seminar provides an opportunity for students to receive feedback from both peers and instructors, helping them identify strengths and areas for improvement.

❖ *Challenges faced in implementation:*

- Due to shyness, some students hesitate to teach.
- Keeping all students engaged throughout the seminar can be challenging, especially if presentations are lengthy or not interactive.

References:

1. <https://effectiviology.com/protege-effect-learn-by-teaching/>
2. https://www.researchgate.net/publication/351905566_Impact_of_Seminars_on_Student_Soft_Skills_Development
3. https://www.ritrjpm.ac.in/images/computer-science/2022-2023/5_VA_CS3362_Learning%20by%20teaching.pdf
4. https://www.ritrjpm.ac.in/images/computer-science/2022-2023/Unit_5_Learning_by_Teaching.pdf

